

Moidul Hasan

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Highlights

- Studied and developed **tuberculosis detection models using deep learning and explainable AI from Chest X-Rays**, improving model interpretability and generalization across datasets through knowledge distillation and localized TB region identification
- Developed a **low-cost smartphone ophthalmoscope**, enhancing retinal screening accessibility with image processing techniques for early glaucoma detection via cup-to-disc ratio calculation

Education

Bangladesh University of Engineering and Technology, Feb 2020 – March 2025
Dhaka, Bangladesh
BSc. in Biomedical Engineering, Faculty of Electrical and Electronic Engineering

- GPA: 3.49/4.0
- **Coursework:** Computer Programming, Embedded Systems and Interfacing, Biomedical Signals and Systems, Digital Signal Processing, Medical Imaging, Healthcare System Management, Telemedicine

Research Interests

Artificial Intelligence in Healthcare, Computer Vision, Biomedical Signal and Image Processing, Clinical Decision Support Systems, mHealth and IoT-Based Healthcare Solutions

Research Experience

Deep Learning Models for Tuberculosis Detection from Chest X-rays Jan 2024 – Present
Undergraduate Thesis; journal manuscript preparation under process

- Improved model focus using knowledge distillation for better interpretability
- Localized TB regions without bounding box information via explainable AI methods
- Enhanced generalization across datasets with interpretable predictions

View-Retina: Low-cost Smartphone Ophthalmoscope July 2022 – Nov 2023
Capstone Project; journal manuscript preparation under process

- Designed a low-cost, portable smartphone ophthalmoscope to enhance retinal screening accessibility
- Implemented image processing techniques to calculate the cup-to-disc ratio (CDR) for early glaucoma detection
- Utilized adaptive thresholding and histogram smoothing for accurate optic disc and cup segmentation

Diabetic Retinopathy Detection July 2022 – Feb 2023
DL Project

- Executed a deep learning multiclass classification model based on a research paper for *referable diabetic retinopathy detection*
- Created a web app for patient image diagnosis, improving clinical accessibility

Significance of Chest X-ray Preprocessing in Pneumonia Detection 2024
Image Processing Project

- Optimized X-Ray contrast through mask generation and edge detection methods
- Improved pneumonia detection with sophisticated image preprocessing techniques

PakhiDroneBD Sept 2024 - Present
Agro-drone Startup; Implementation Stage

- Co-Founder & Chief Research Officer (CRO)
- Developing an AI-assisted agro-drone to reduce pesticide waste and chemical exposure via aerial spraying

Achievements

- University Innovation Hub Program (UIHP) Cohort 2024
- Healthcare Case Competition Winner, BME Fest 2.0 2024
- Bangladesh Biology Olympiad (BDBO) Best Enzyme Award, Dhaka South 2022

Technologies

Languages: Python, C, MATLAB

Python Packages: Numpy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch

Software & Tools: Solidworks, AutoCAD, Ansys Fluent, COMSOL, LaTeX, MS Office (Words, Excel, PowerPoint)

Internship

GETWELL Pharma Limited, Pran-RFL Group May 2023

Voluntary Works

Bangladesh Biology Olympiad (BDBO) Committee, Dhaka South Dec 2021 - March 2025
Assistant Organizing Secretary

BUET Biomedical Engineering Society (BMES) Student Chapter Dec 2021 - March 2025
Vice President

BUET Animata Nov 2023 - March 2025
Co-Founder & Vice President (Communications)

BUET ACS Student Chapter Jan 2024 - March 2025
Event Management Chair

References

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